

AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

PHASE 2: REQUIREMENT ANALYSIS

1. Comprehensive Business Requirements Analysis

The requirement analysis phase translates institutional pain points identified during ideation into concrete, enforceable functional and technical requirements. The system must provide automated, deterministic classification and communication workflows while maintaining data consistency across the ServiceNow database schema.

- **BR-01 (Automated Intake & Classification):** The system must automatically analyze incoming ticket descriptions upon submission and assign appropriate classification tags without human intervention.
- **BR-02 (Hierarchical Dependent Mapping):** The system must support two-tier classification (Category and Subcategory) and enforce dependent validation to prevent impossible combinations (e.g., Category: Network with Subcategory: Projector).
- **BR-03 (Automated Caller Acknowledgment):** The system must send a personalized transactional email to the caller immediately upon ticket generation containing the ticket reference number.
- **BR-04 (Zero-Script Maintenance):** The platform configuration must rely strictly on ServiceNow native no-code capabilities (Flow Designer, Dictionary Overrides, Dependent Choice lists) to facilitate long-term maintenance by junior support staff.

2. Detailed Functional Requirements (FR)

Functional requirements outline the exact runtime behavior expected from the custom ServiceNow table and automation engine across all operational scenarios.

Requirement ID	Functional Description	Trigger / Condition	System Response / Output	Priority
FR-01	Trigger Automation Engine	New record inserted into Incident WorkFlow table with Category is Empty	Instantiate 'Auto Classify School IT Tickets' Flow execution	Critical
FR-02	Network Keyword Classification	Short Description contains 'Wi-Fi' or 'Network'	Update Record: Category = Network, Subcategory = Wi-Fi	High
FR-03	Hardware Keyword Classification	Short Description contains 'Projector' or 'Hardware'	Update Record: Category = Hardware, Subcategory = Projector	High
FR-04	Access Keyword Classification	Short Description contains 'Forgot password' or 'Password'	Update Record: Category = Access, Subcategory = Forgot Password	High
FR-05	Performance Keyword Classification	Short Description contains 'Slow computer' or 'Slow'	Update Record: Category = Performance, Subcategory = Slow Computer	High
FR-06	Outbound Notification Generation	Successful categorization update of incident record	Dispatch email to caller's email address	High

3. Non-Functional Requirements & Constraints (NFR)

Non-functional requirements govern system performance, operational integrity, data security, and platform scalability.

- **NFR-01 — Performance & Execution Speed:** Automated classification and outbound email creation must complete execution within 2.0 seconds from the database commit timestamp.
- **NFR-02 — Data Integrity & Consistency:** Auto-numbering must guarantee unique sequential ticket identification without collisions or numbering gaps.
- **NFR-03 — Security & Role-Based Access Control (RBAC):** Only authenticated school users (students, teachers, administrators) may insert records, while only IT fulfillers may alter ticket states.
- **NFR-04 — Portability & Scope Management:** All platform customizations (tables, fields, choices, dictionary entries, flow logic) must be captured cleanly within a dedicated Global Update Set for migration.

4. Stakeholder Requirement Traceability Matrix

To verify complete stakeholder alignment, the matrix below connects user roles to concrete functional deliverables.

Stakeholder Group	Primary Need	Mapped Feature / Deliverable	Verification Metric
Students & Faculty	Rapid ticket submission with minimal typing	Simplified incident intake form (only Caller & Short Description required)	User submission time < 30 seconds
Students & Faculty	Instant confirmation of ticket receipt	Flow Designer automated outbound email notification	Email dispatch time < 5 seconds

Tier-1 Service Desk Agents	Clean, pre-sorted queue without manual sorting	Automated Category & Subcategory population via Flow Designer	Manual categorization volume = 0%
IT Operations Management	Accurate reporting metrics by incident category	Dependent choice dictionary configuration	Category mismatch errors = 0%